

Asymptotic behavior of partial sums

François Roueff

<http://perso.telecom-paristech.fr/~roueff/>

Telecom Paris

M2 SFA

Outline

- 1 Determining the integration order
 - Differences and partial sums of a time series
 - Application : unit root test
 - Application: testing a change in the mean
- 2 Complements
 - Convergence in D
 - Invariance principle
- 3 An illustrative example with R

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Example : USD vs EUR currency exchange rate

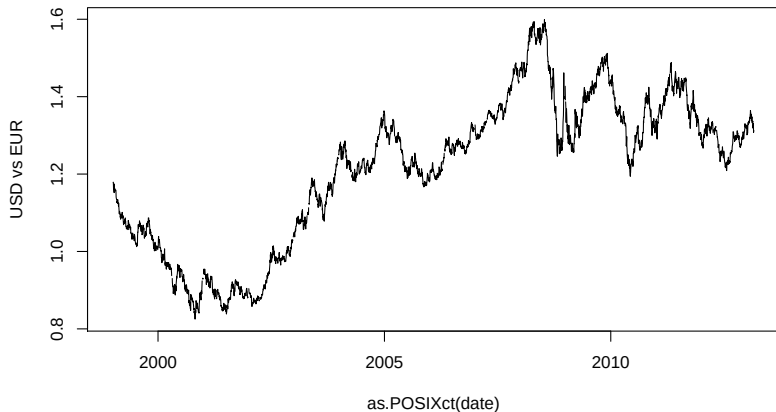


Figure: Daily currency exchange rate : price of 1 Euro in US Dollars.

Example : USD vs EUR currency exchange rate (cont.)

To use a **stationary** model, one applies the **differencing operator**

$$Y = \Delta X \quad \text{defined by} \quad Y_t = X_t - X_{t-1}, \quad t \in \mathbb{Z}.$$

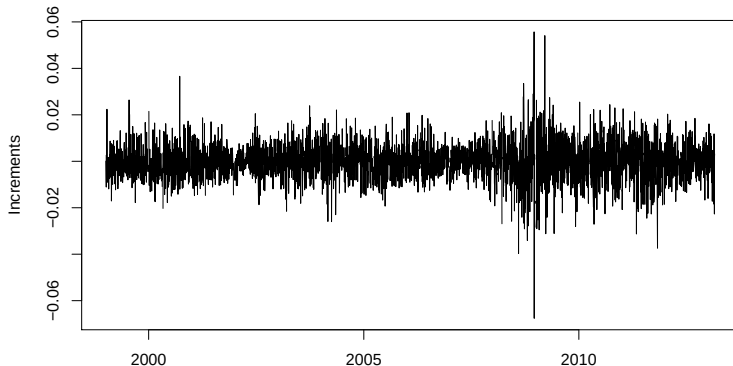
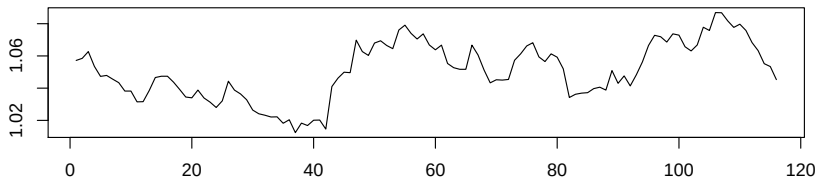


Figure: Increments of daily USD-EUR currency exchange rate.

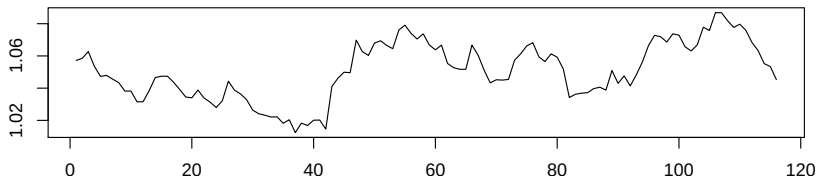
Questions

- ▷ Is it always obvious when to difference the data or not?



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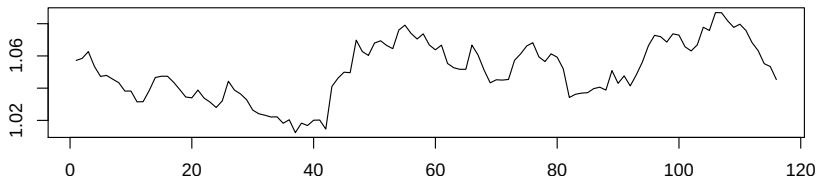
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Definition : Integration order

Let $d = 1, 2, \dots$, we say that Y is an $I(d)$ process (*integrated of order d*) if $\Delta^d Y$ is stationary *up to a polynomial trend* while $\Delta^{d-1} Y$ is not.

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- ▶ Assume that X is $I(d)$ for some unknown $d \geq 0$ with polynomial trend of degree at most d .

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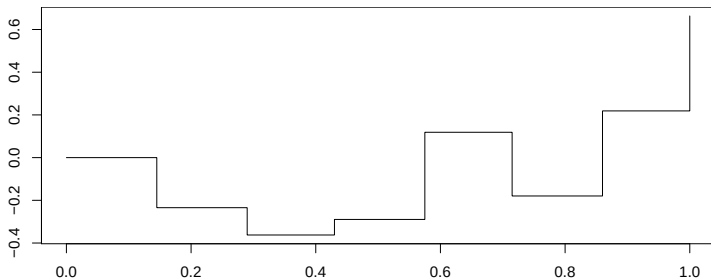
Determine the integration order

- ▶ Assume that X is $I(d)$ for some unknown $d \geq 0$ with polynomial trend of degree at most d .
- ▶ Find out d such that $X = \Delta^d Y$ is an $I(0)$ process with polynomial trend of degree at most d .
- ▶ Equivalently X is stationary and cannot be written as ΔZ with Z stationary.

Integrating a time series : partial sums

Consider a sample $X_1, \dots, X_n$. Define

$$S_n(t) = \sum_{k=1}^{\lfloor tn \rfloor} X_k \quad \text{on } t \in [0, 1] .$$



Then S_n is a **piecewise constant** function.

Large sample behavior of $S_n(t)$

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$$S_n(t) = Z_{[nt]} - Z_0 .$$

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For a fixed $t \in (0, 1]$, we have

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For a fixed $t \in (0, 1]$, we have

$$S_n(t) = [nt] \hat{\mu}_{[nt]} .$$

Hence for a “standard” weakly stationary process X with mean μ and spectral density f , we expect that

$$S_n(t) \sim t\mu n \quad \text{a.s.}$$

Centering

To get rid of the mean, we shall consider

$$S_n(t) - tS_n(1) = \sum_{k=1}^{[tn]} (X_k - \mu) - t \sum_{k=1}^n (X_k - \mu) + (tn - [tn])\mu$$

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where

$$\bar{S}_n(t) = \sum_{k=1}^{[tn]} (X_k - \mu).$$

Under many stationary models for X , the empirical mean is **asymptotically normal**, and we have the weak convergence

$$\frac{1}{\sqrt{n}} \bar{S}_n(t) = \frac{\sqrt{[tn]}}{\sqrt{n}} \frac{1}{\sqrt{[tn]}} \sum_{k=1}^{[tn]} (X_k - \mu) \Rightarrow \mathcal{N}(0, 2\pi t f(0)).$$

Convergence at a fixed t

We immediately get that

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To get the weak convergence of $n^{-1/2} (S_n(t) - tS_n(1))$, we need the **joint convergence** of $n^{-1/2}(\overline{S}_n(t), \overline{S}_n(1))$.

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In fact, we may directly get the **fidi** convergence of

$$n^{-1/2} (S_n(t) - tS_n(1)) , \quad t \in [0, 1],$$

from that of

$$n^{-1/2} \overline{S}_n(t), \quad t \in [0, 1] .$$

Fidi convergence

Assumption on X

Let $X = \mu + F_\psi(Z)$ be a linear process with short memory and assume that $Z \sim \text{IID}(0, \sigma^2)$.

Let $0 < t_1 < t_2 < \dots < t_p \leq 1$.

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Let $0 < t_1 < t_2 < \dots < t_p \leq 1$.

$$X_1, X_2, \dots, X_{[t_1 n]}, \quad X_{[t_1 n]+1}, \dots, X_{[t_2 n]}, \quad \dots, \quad X_{[t_{p-1} n]+1}, \dots, X_{[t_p n]}$$

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||

$$[t_1 n](\hat{\mu}_{[t_1 n]} - \mu) \quad \dots \quad \dots \quad \dots$$

Elaborating on the proof of the asymptotic normality of $\hat{\mu}_n$, it may be shown that these increments are asymptotically normal and independent.

Fidi convergence (cont.)

We get that

$$\frac{1}{\sqrt{n}} \begin{bmatrix} \overline{S}_n(t_1) \\ \overline{S}_n(t_2) - \overline{S}_n(t_1) \\ \vdots \\ \overline{S}_n(t_p) - \overline{S}_n(t_{p-1}) \end{bmatrix} \Rightarrow \mathcal{N} \left(0, 2\pi \textcolor{red}{f}(0) \begin{bmatrix} t_1 & 0 & \dots & 0 \\ 0 & t_2 - t_1 & \ddots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & t_p - t_{p-1} \end{bmatrix} \right)$$

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Consequence

$$\frac{1}{\sqrt{n}} \overline{S}_n \xrightarrow{\text{fidi}} \sqrt{2\pi \textcolor{red}{f}(0)} B,$$

where $(B(t))_{t \in [0,1]}$ is the **standard Brownian motion** defined on $[0, 1]$, that is, a **Gaussian process** with **stationary and independent** increments.

Fidi convergence : limitations

The fidi convergence allows to derive the limit of

$$g\left(\frac{1}{\sqrt{n}} \overline{S}_n(t_i), i = 1, \dots, p\right)$$

for all $0 \leq t_1 < t_2 < \dots < t_p \leq 1$ and every continuous $g : \mathbb{R}^p \rightarrow \mathbb{R}$.

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However, we cannot say anything about

$$\sup_{t \in [0,1]} \overline{S}_n(t), \quad \int_0^1 \overline{S}_n(t) dt, \quad \dots$$

Convergence in a functional metric space

Question

Do we have a better adapted weak convergence for this sequence of random $[0, 1] \rightarrow \mathbb{R}$ functions ?

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Do we have a better adapted weak convergence for this sequence of **random** $[0, 1] \rightarrow \mathbb{R}$ functions ?

Quick answer

One can actually prove that

$$\frac{1}{\sqrt{n}} \overline{S}_n \Longrightarrow \sqrt{2\pi f(0)} B \quad \text{in } (D, d),$$

where (D, d) is a metric metric space which makes the sup continuous, see Section 1.

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Hypotheses

Simplified model for the stationary VS integrated process problem.

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Suppose you observe a sample $X_1, \dots, X_n$ and you want to distinguish between

H_0 : X is a pure random walk ,

$$X_t = \sum_{k=1}^t \epsilon_k, \quad \text{where } \epsilon \sim \text{IID}(0, \sigma^2) ,$$

against

H_1 : X is a stationary AR(1) process,

$$X_t = \phi X_{t-1} + \epsilon_t, \quad \text{where } |\phi| < 1, \epsilon \sim \text{IID}(0, \sigma^2) .$$

Estimation of ϕ

Note that, under H_0 , we have

$$X_t = \phi X_{t-1} + \epsilon_t, \quad \text{for all } t = 2, \dots, n \quad \text{with } \phi = 1.$$

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$$\hat{\phi}_n = \operatorname{argmin}_{a \in \mathbb{R}} \sum_{t=2}^n (X_t - a X_{t-1})^2$$

(1)

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$$\begin{aligned} \hat{\phi}_n &= \underset{a \in \mathbb{R}}{\operatorname{argmin}} \sum_{t=2}^n (X_t - a X_{t-1})^2 \\ &= \frac{\sum_{t=2}^n X_{t-1} X_t}{\sum_{t=2}^n X_{t-1}^2} \\ &= \phi + \frac{\sum_{t=2}^n X_{t-1} \epsilon_t}{\sum_{t=2}^n X_{t-1}^2}. \end{aligned} \tag{1}$$

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Then, we get that, under H_1 ,

$$\hat{\phi}_n = \phi + O_p(n^{-1/2}).$$

Expressions using partial sums $S_n(t)$

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Then we have, under H_0 ,

$$\frac{1}{n} \sum_{t=2}^n X_{t-1}^2 = \sum_{t=2}^n \int_{(t-1)/n}^{t/n} S_n^2(u) \, du = \int_0^1 S_n^2(u) \, du . \quad (2)$$

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On the other hand, under H_0 ,

$$2 \sum_{t=2}^n X_{t-1} \epsilon_t = 2 \sum_{1 \leq s < t \leq n} \epsilon_s \epsilon_t = \left(\sum_{t=1}^n \epsilon_t \right)^2 - \sum_{t=1}^n \epsilon_t^2 ,$$

and thus,

$$\frac{1}{n} \sum_{t=2}^n X_{t-1} \epsilon_t = \frac{1}{2} (n^{-1/2} S_n(1))^2 - \frac{1}{2n} \sum_{t=1}^n \epsilon_t^2 . \quad (3)$$

Asymptotic behavior

Since $\epsilon \sim \text{IID}(0, \sigma^2)$, we have

$$\frac{1}{2n} \sum_{t=1}^n \epsilon_t^2 \xrightarrow{\text{a.s.}} \frac{\sigma^2}{2},$$

and the Donsker theorem gives that

$$\frac{1}{\sqrt{n}} S_n \Rightarrow \sigma B \quad \text{in } (D, d).$$

With (1), (2) and (3), we finally get that, under H_0 ,

$$n \left(\hat{\phi}_n - 1 \right) \Rightarrow \frac{B^2(1) - 1}{2 \int_0^1 B^2(u) du},$$

whose distribution is called the **Dickey-Fuller distribution**.

Unit root test

Define the test statistic

$$T_n = n \left| \hat{\phi}_n - 1 \right| ,$$

so that, under H_0 , T_n has a known asymptotic distribution,

$$T_n \Rightarrow \frac{|B^2(1) - 1|}{2 \int_0^1 B^2(u) \, du} ,$$

while, under H_1 ,

$$T_n = n|\phi - 1| + O_p(\sqrt{n}) .$$

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Hypotheses

Suppose you observe a sample $X_1, \dots, X_n$ such that

$$X_t = \mu(t) + \epsilon_t ,$$

where $\epsilon \sim \text{IID}(0, 1)$, and you want to distinguish between

H_0 : X has a constant mean,

$$\mu(t) = \mu \quad \text{for all } t .$$

against

H_1 : X has a change in the mean,

$$\mu(t) = \begin{cases} \mu_1 & \text{for all } t \leq un \\ \mu_2 & \text{for all } t > un \end{cases} , \quad \text{where } \mu_1 \neq \mu_2, u \in (0, 1) .$$

Note that u is defined in **rescaled time**.

Testing a change in the mean (cont.)

Define the statistic

$$T_n = \sup_{t \in [0,1]} \frac{1}{\sqrt{n}} |S_n(t) - tS_n(1)| .$$

Testing a change in the mean (cont.)

Define the statistic

$$T_n = \sup_{t \in [0,1]} \frac{1}{\sqrt{n}} |S_n(t) - tS_n(1)| .$$

Then under H_0 , we have

$$T_n \Rightarrow \sup_{t \in [0,1]} |B(t) - tB(1)| .$$

Meanwhile, under H_1 , we have

$$T_n \sim |\mu_1 - \mu_2| u(1-u) \sqrt{n} \quad \text{a.s.}$$

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Unfortunately, D is not **separable** when endowed with the **sup norm**.

Definition : Skorohod topology

Let $x, y \in D$. Define

$$d(x, y) = \inf_{\lambda} (\|x \circ \lambda - y\|_{\infty} + \|\lambda - \text{Id}\|_{\infty}) ,$$

where $\text{Id}(t) = t$ and the $\inf$ is taken over all continuously increasing $[0, 1] \rightarrow [0, 1]$ bijections. Then (D, d) is a **separable metric space**. Moreover the sup is **continuous** on D .

Tightness

Definition : tightness

A sequence (X_n) of random elements in D is said to be **tight** in (D, d) if, for any $\epsilon > 0$, there exists a **compact set** $K \subset D$ such that

$$\sup_n \mathbb{P}(X_n \notin K) \leq \epsilon .$$

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Prohorov's Theorem

If (X_n) is tight, one can extract a subsequence (X_{α_n}) which weakly converges in (D, d) .

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Consequence

If (X_n) is tight in (D, d) and $X_n \xrightarrow{\text{fidi}} X$, where X is a random element of D , then (X_n) weakly converges to X in (D, d) .

A Donsker theorem

Theorem

Let $X = \mu + F_\psi(Z)$ be a linear process with short memory and assume that $Z \sim \text{IID}(0, \sigma^2)$ with $\mathbb{E}[Z_0^4] < \infty$. Then we have

$$\frac{1}{\sqrt{n}} \bar{S}_n \Rightarrow \sqrt{2\pi f(0)} B \quad \text{in } (D, d),$$

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where B is the continuous standard Brownian motion defined on $[0, 1]$.

Consequence

Thus we have

$$\frac{1}{\sqrt{n}} (S_n(t) - tS_n(1)) \Rightarrow \sqrt{2\pi f(0)} (B - B(1) \text{Id}).$$

The process $B - B(1) \text{Id}$ is called a Brownian bridge.

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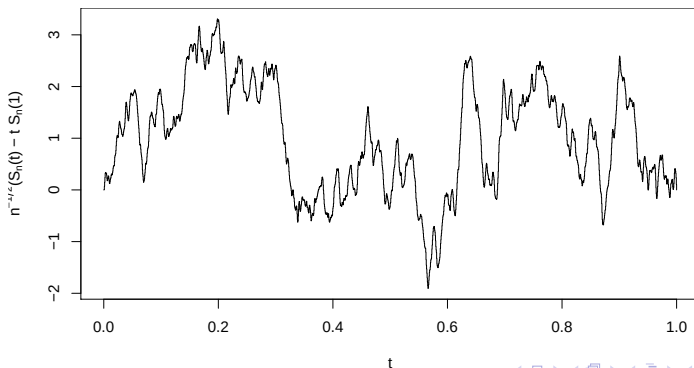
Invariance principle

The Donsker theorem

$$\frac{1}{\sqrt{n}} \bar{S}_n \Rightarrow \sqrt{2\pi f(0)} B \quad \text{in } (D, d),$$

is sometimes called an **invariance principle**, since the limit process is the same (up to a multiplicative constant)

for a **large class** of **stationary** processes X .



Does the invariance principle always hold for stationary processes ?

What happens if we try to enlarge this class to

- ▷ Non-linear processes ?
- ▷ Infinite variance processes ?
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Quick answers

- ▷ A large class of non-linear processes satisfies the same Donsker theorem ($\rightarrow$ **Markov** processes).

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- ▷ Long memory processes ?

Quick answers

- ▷ A large class of non-linear processes satisfies the same Donsker theorem ($\rightarrow$ Markov processes).
- ▷ For a large class of processes X with infinite variance, the limit is not Gaussian and depends on the extremal behavior of X ($\rightarrow$ regularly varying processes).

Does the invariance principle always hold for stationary processes ?

What happens if we try to enlarge this class to

- ▷ Non-linear processes ?
- ▷ Infinite variance processes ?
- ▷ Long memory processes ?

Quick answers

- ▷ A large class of non-linear processes satisfies the same Donsker theorem ($\rightarrow$ Markov processes).
- ▷ For a large class of processes X with infinite variance, the limit is not Gaussian and depends on the extremal behavior of X ($\rightarrow$ regularly varying processes).
- ▷ For a large class of linear long memory processes X , the limit is Gaussian but is not a Brownian motion ($\rightarrow$ self-similar processes).

Example : Nile river data

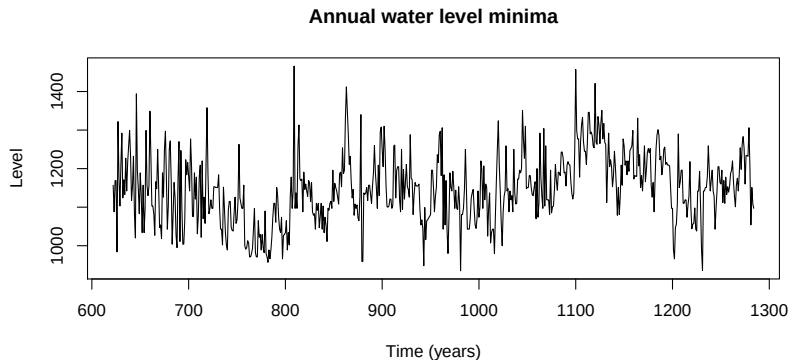


Figure: Annual WATER LEVEL minima of the Nile river studied by H. Hurst (1951).

Example : Nile river data (cont.)

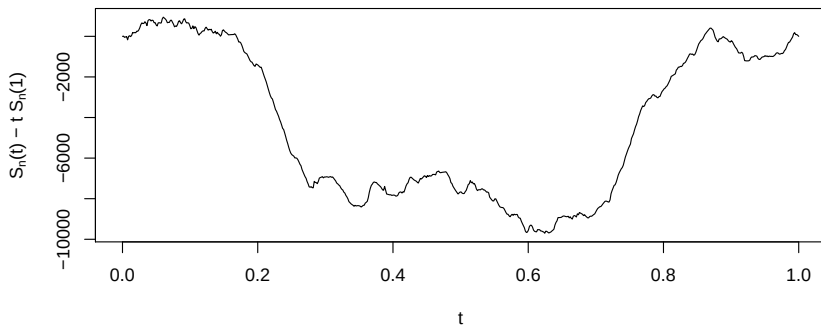
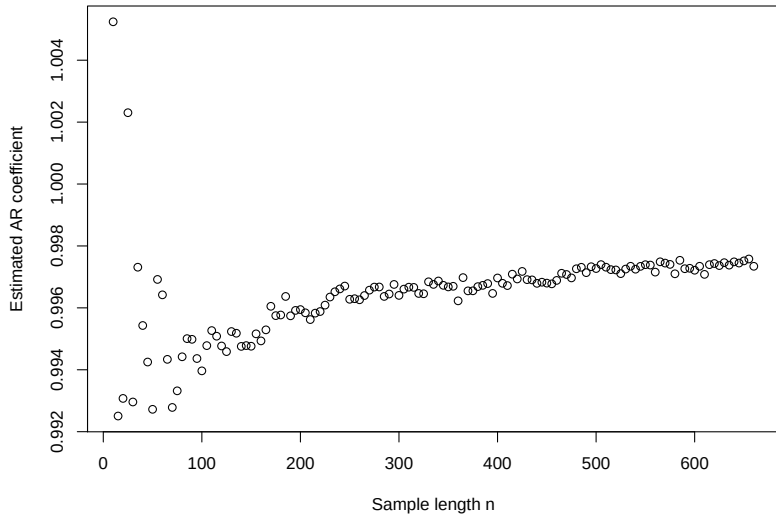


Figure: Integration of the Nile river time series ($S_n(t) - tS_n(1)$ for $t \in [0, 1]$).

Example : Nile river data (cont.)



Some hints about long memory

Let $Z \sim \text{IID}(0, \sigma^2)$. Define, for some $d \in (0, 1/2)$,

$$X = \widehat{F}_{g_d}(Z) \quad \text{with} \quad g_d(\lambda) = (1 - e^{-i\lambda})^{-d}.$$

In other words, we set

$$X_t = \int_{\mathbb{T}} e^{i\lambda t} (1 - e^{-i\lambda})^{-d} d\widehat{Z}(\lambda)$$

It is a **long memory linear** process with **long memory parameter** d because its spectral density satisfies

$$f(\lambda) = \frac{\sigma^2}{2\pi} |g_d(\lambda)|^2 \sim \frac{\sigma^2}{2\pi} |\lambda|^{-2d} \quad \text{as } \lambda \rightarrow 0.$$

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What is the asymptotic behavior of S_n as $n \rightarrow \infty$?

Some hints about long memory (cont)

Simplifying Assumption

Suppose that Z is Gaussian.

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To get the **functional** convergence, we only need to find out the asymptotic behavior of the autocovariance function of S_n .

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Suppose that Z is Gaussian.

Then X and S_n are centered Gaussian processes.

To get the fidi convergence, we only need to find out the asymptotic behavior of the autocovariance function of S_n .

Observe that, for all $s, t \in [0, 1]$,

$$\text{Cov}(S_n(s), S_n(t)) = \frac{1}{2} [\text{Var}(S_n(t)) + \text{Var}(S_n(s)) - \text{Var}(S_n(t) - S_n(s))] .$$

Hence we only need to study

$$\text{Var} \left(\sum_{k=1}^m X_k \right) \quad \text{as } m \rightarrow \infty .$$

Some hints about long memory (cont)

We have, for all $t \in (0, 1]$,

$$\begin{aligned}\text{Var} \left(\sum_{k=1}^m X_k \right) &= \frac{\sigma^2}{2\pi} \int_{\mathbb{T}} \left| \sum_{k=1}^m e^{-i\lambda k} \right|^2 \left| 1 - e^{-i\lambda} \right|^{-2d} d\lambda \\ &= \frac{\sigma^2}{\pi} \int_0^\pi \sin^2(m\lambda/2) \underbrace{\frac{|1 - e^{-i\lambda}|^{-2d}}{\sin^2(\lambda/2)}}_{4\lambda^{-2d-2} + O(\lambda^{-2d-1})} d\lambda \\ &= \frac{4\sigma^2}{\pi} m^{2d+1} \int_0^{m\pi} \sin^2(u/2) u^{-2d-2} du + O(\dots) \\ &\sim c m^{2d+1} \quad \text{as } m \rightarrow \infty ,\end{aligned}$$

where

$$c^2 = \frac{4\sigma^2}{\pi} \int_0^\infty \sin^2(u/2) u^{-2d-2} du > 0 .$$

Some hints about long memory (cont)

We obtain that for all $0 \leq s < t \leq 1$,

$$\lim_{n \rightarrow \infty} n^{-2d-1} \text{Var} (S_n(t) - S_n(s)) = c^2 |t - s|^{2d+1} .$$

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Consequence

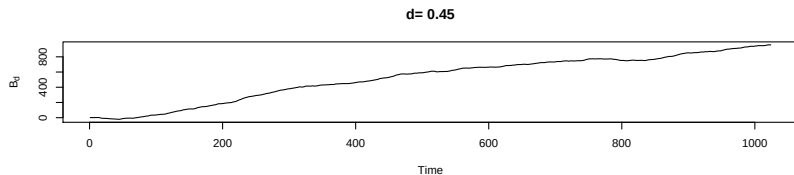
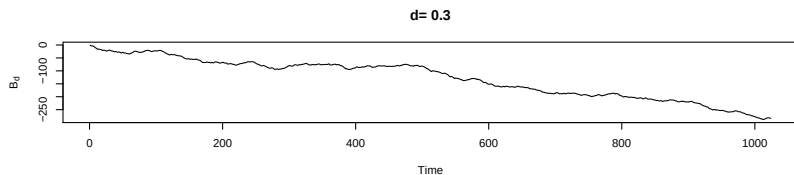
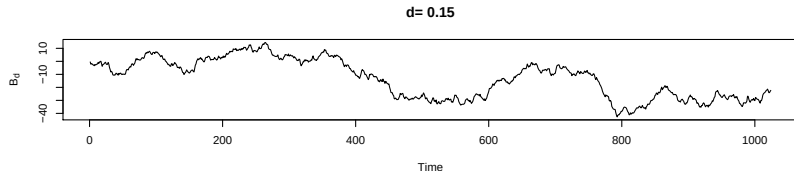
Under the above assumptions, we have, as $n \rightarrow \infty$,

$$n^{-d-1/2} S_n \xrightarrow{\text{fdi}} c B_d ,$$

where $B_d(t)$ is a Gaussian process defined on $t \in [0, 1]$ such that $B_d(0) = 0$ and, for all $0 \leq s < t \leq 1$,

$$\text{Var} (B_d(t) - B_d(s)) = |t - s|^{2d+1} .$$

Samples paths of the limit



Concluding remarks

- ▶ Note that the rate of convergence of $\hat{\mu}_n = n^{-1}S_n(1)$ is $n^{d-1/2}$, which is slower than the usual $n^{-1/2}$.

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- ▷ The process B_d is called a **fractional Brownian motion** with **Hurst index** $H = d + 1/2$.

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- ▷ The process B_d is called a **fractional Brownian motion** with **Hurst index** $H = d + 1/2$.
- ▷ It can be shown that B_d is a continuous process and

$$n^{-d-1/2}S_n \Rightarrow c B_d \quad \text{in } (D, d) .$$

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- ▶ **Statistical inference** : how to estimate the parameter d , how to distinguish stationarity/long memory/non-stationarity ?

- 1 Determining the integration order
- 2 Complements
- 3 An illustrative example with R

```
#####
#           Examples Donsker and applications           #
#####
set.seed(1)
# wait after each plot
dx <- 0.01 # seconds
y <- arima.sim(n = 2^11,
               model=list(ar=0.7, ma=0.9), sd=1)
dev.new(width=10, height=5)
# plot of original time series
ts.plot(y)
op <- par(mar=c(3,2,0,1))
for (n in seq(20,length(y),by=10)) {
  t <- seq(from=0,to=1,by=1/length(y))
  x <- c(0,cumsum(y[1:n]))/sqrt(n)
  s <- x[1+floor(n*t)] - t*x[n+1]
  op <- par(mar=c(2,2,0,1))
  plot(t,s,type='s',ylab='',xlab='',
        xlim=c(0,1),ylim=c(-4,4))
  Sys.sleep(dx)
}
#####
#           unit root test                               #
#####
# Clear variables and graphics
rm(list=ls())
# setting
# test statistic function
tnstat <- function(y){
  n <- length(y)
  hphi <- sum(y[2:n]*y[1:n-1])/sum((y[1:n-1])^2)
  T <- n*abs(hphi-1)
  return(list(stat=T,hphi=hphi))
}
nmax <- 2^10;tphi <- 0.95;set.seed(1)
heps <- rnorm(nmax,mean=0,sd=1)
```

```

x1 <- arima.sim(n = nmax, list(ar = tphi), innov=hyps)
x0 <- cumsum(hyps); n <- 2^8
ts.plot(as.ts(x0[1:n]),as.ts(x1[1:n]),
        ylab='',col=c(1,2))
#initiate
t0 <- c();t1 <- c();hphi <- c()
nseq <- seq(10,nmax,by=10)
for (n in nseq){
  t0 <- c(t0,tnstat(x0[1:n])$stat)
  t1 <- c(t1,tnstat(x1[1:n])$stat)
  hphi <- c(hphi,tnstat(x1[1:n])$hphi)
}
# plot the test statistic behavior as n increases
plot(nseq,t1,type='o',col=2,xlab='Sample length n',
     ylab=expression(T[n]))
points(nseq,t0,type='o',col=1)
# plot estimation of phi
plot(nseq,hphi,xlab='Sample length n',
     ylab="Estimated AR coefficient")
#####
#           change in the mean           #
#####
# Clear variables and graphics
rm(list=ls())
# test statistic function
tnstat <- function(y){
  n <- length(y)
  S <- cumsum(y)
  Sc <- abs(S-S[n]*(1:n)/n)/sqrt(n)
  T <- max(Sc)
  loc <- which.max(Sc)
  return(list(stat=T,loc=loc))
}
# setting
nmax <- 2^12; mu <- 1.0; mu1 <- .8
mu2 <- 1.2; tu <- 0.7; set.seed(1)

```

```

heps <- rnorm(nmax,mean=0,sd=1)
n <- 2^8; tk <- floor(tu*n); y0 <- mu+heps[1:n]
y1 <- c(rep(mu1,tk),rep(mu2,n-tk))+heps[1:n]
# plot n first samples
ts.plot(as.ts(y0),as.ts(y1[1:tk]),
        ts(y1[tk+1:n],start=tk+1),
        col=c(1,2,3),xlim=c(0,n),ylab="")
#initiate
t0 <- c();t1 <- c();ct <- c()
nseq <- seq(10,nmax,by=10)
for (n in nseq){
  y0 <- mu+heps[1:n]
  y1 <- c(rep(mu1,floor(tu*n)),
          rep(mu2,n-floor(tu*n)))+heps[1:n]
  t0 <- c(t0,tnstat(y0)$stat)
  tmp1 <- tnstat(y1)
  t1 <- c(t1,tmp1$stat)
  ct <- c(ct,tmp1$loc/n)
}
# plot the test statistic behavior as n increases
plot(nseq,t1,type='o',col=2,xlab='Sample length n',
     ylab=expression(T[n]))
points(nseq,t0,type='o',col=1)
# plot estimation of change time
plot(nseq,ct,xlab='Sample length n',
     ylab="Estimated position")

```